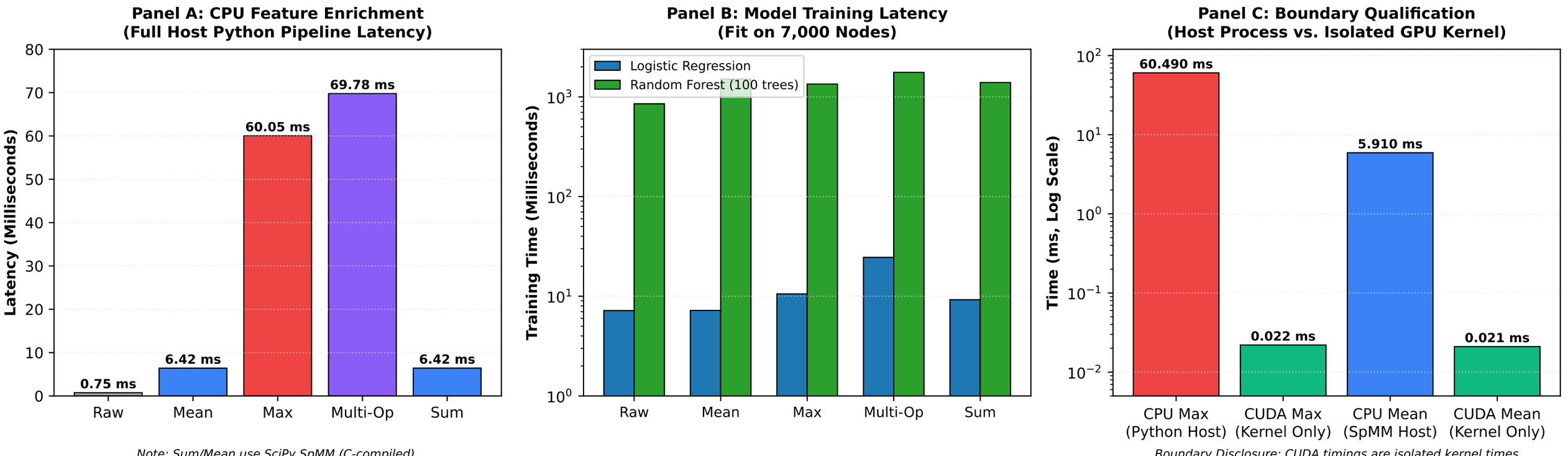
Computational Timing Breakdown Across Pipeline Components
Panel A: CPU Feature Enrichment
(Full Host Python Pipeline Latency)
Latency (Milliseconds)
0
10
20
30
40
50
60
70
80
0.75 ms
6.42 ms
60.05 ms
69.78 ms
6.42 ms
Raw
Mean
Max
Multi-Op
Sum
Note: Sum/Mean use SciPy SpMM (C-compiled).
Max uses per-node Python reduction loop.
Panel B: Model Training Latency
(Fit on 7,000 Nodes)
Training Time (Milliseconds)
Logistic Regression
Random Forest (100 trees)
Raw
Mean
Max
Multi-Op
Sum
Panel C: Boundary Qualification
(Host Process vs. Isolated GPU Kernel)
Time (ms, Log Scale)
60.490 ms
0.022 ms
5.910 ms
0.021 ms
CPU Max
(Python Host)
CUDA Max
(Kernel Only)
CPU Mean
(SpMM Host)
CUDA Mean
(Kernel Only)
Boundary Disclosure: CUDA timings are isolated kernel times
(cudaEventElapsedTime). Not an end-to-end speedup comparison.